Labour Supply 2: Labour Supply and Public Policy

EC306- Wages and Employment

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Goals of This Section

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- Discuss various public policies that affect labour supply
  - Taxes/subsidies
  - Transfers
  - Employment Insurance
  - Worker Compensation
- Show effect of these policies on work decisions
- Examine statistical evidence for the model predictions

Income Maintenance Programs

Overview of Income Maintenance Programs

- Represent about 10% of GDP in Canada
- Aim to
  - Raise income of specific groups of people (e.g. parents, elderly)
  - Supplement low wages
- Include child benefits, employment insurance, social assistance, pensions
- Key challenge is balancing income support with work incentives

Government Transfers in Canada

- In 2017, 83% of families received some form of government transfer
- Major programs:
  - Federal Child Benefits (20.2%) – avg. $6,314
  - Employment Insurance (14.0%) – avg. $8,352
  - Social Assistance (11.3%) – avg. $8,367
  - Old Age Security (25.1%) – avg. $8,566
  - Canada/Quebec Pension Plan (32.4%) – avg. $10,092

Issues in Program Design

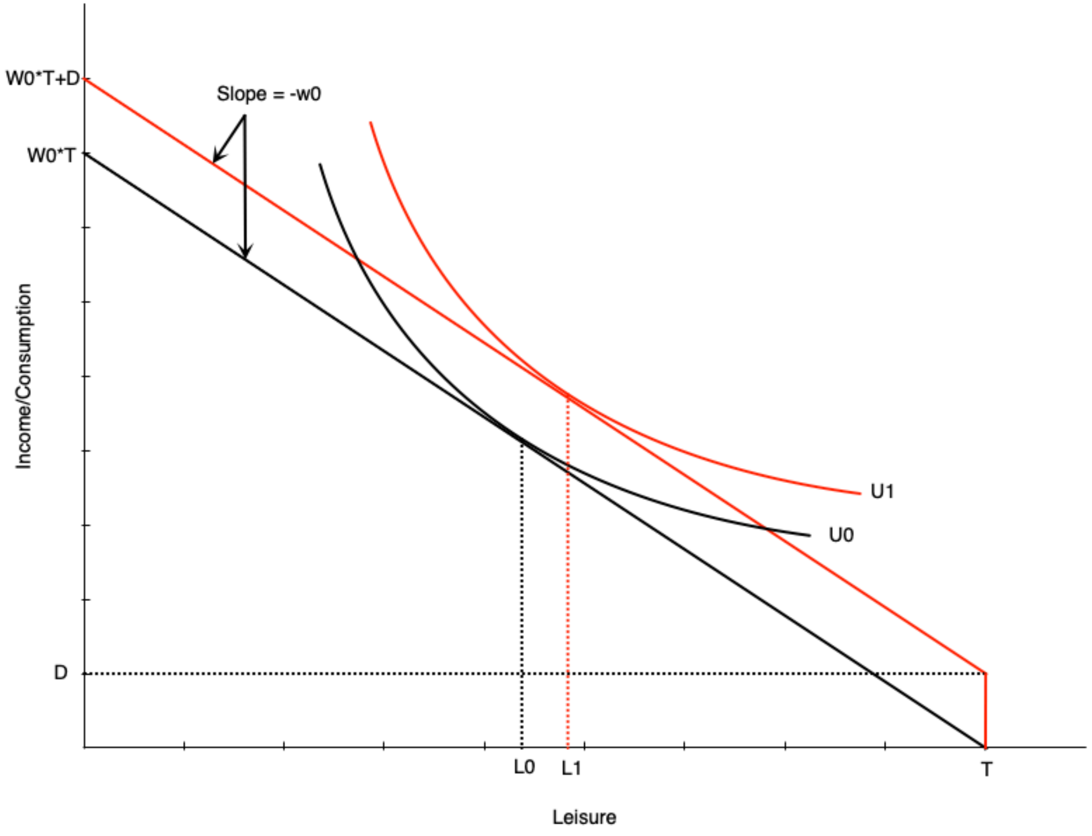
- Universal vs. Targeted Programs
  - Universal programs easy to implement, but expensive and flows to people not in need
  - Targeted programs benefit only those in need, but can be costly to implement
- Work Disincentives
  - Some programs that support income reduce the incentive to work
  - Example: a welfare benefit that increases non-work income
    - Can encourage people with preference for leisure to drop out of labour force
- Permanent vs. transitory income loss
  - Need different design for each situation
  - Temporary layoff requires short-term support
  - Long-term disability requires longer support and potentially more costs to reintegrate into workforce

Demogrants, Welfare, Wage Subsidies

Introduction

- We first examine standard income support programs
- Demogrants
  - A fixed payment to all individuals regardless of income or employment status
- Welfare
  - Lump-sum payment for non-workers, reduced dollar for dollar with earned income
- Negative Income Tax (NIT)
  - A welfare payment that decreases at a lower rate than dollar for dollar with earned income
- Wage Subsidies
  - An addition to the wage rate

Demogrant



- Demogrant is an unconditional lump sum income transfer
- Shifts the budget line up by the amount D
- Because wage does not change, this is a pure income effect
- If leisure is a normal good, optimal leisure falls
- Unambiguously negative work incentive effects
  - Leads to lower work hours
  - Can also lead to non-participation

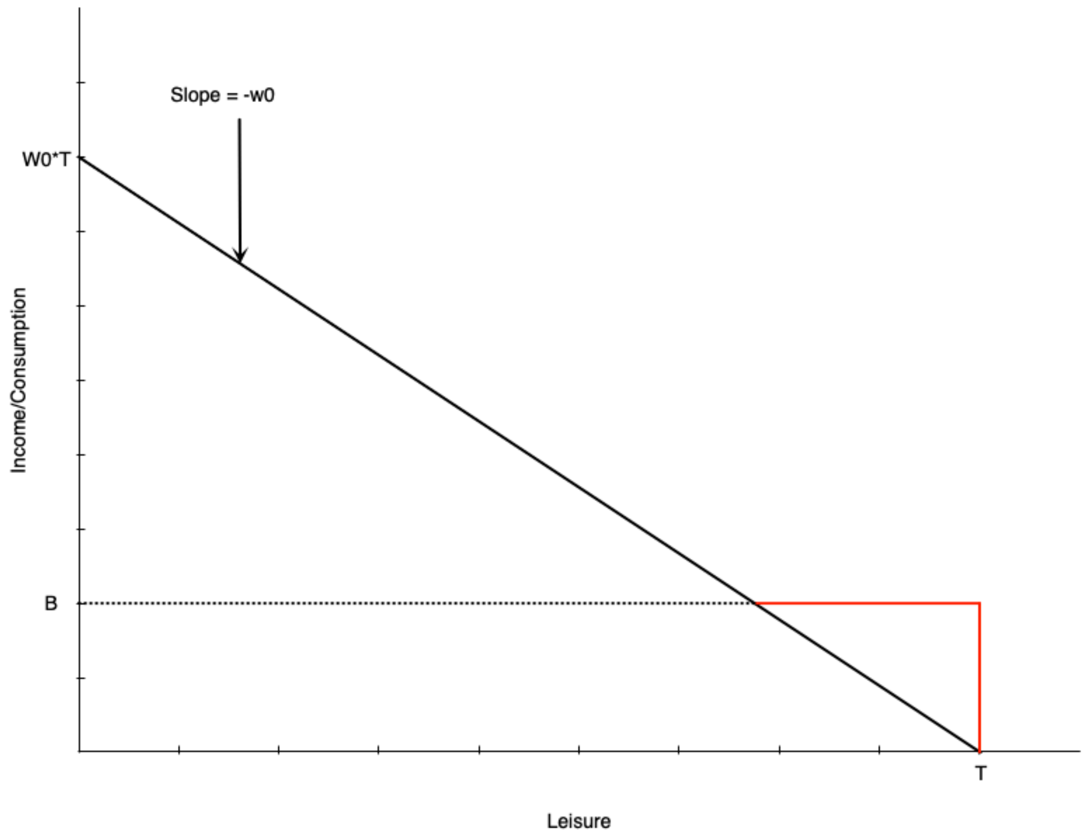
Demogrant

- Universal Child Care Benefit (UCCB) is a Canadian policy similar to a demogrant
  - Paid to families with children under 18
  - $160/month for each child under 6
  - $60/month for each child aged 6-17
  - Canada Child Benefit (CCB) replaced UCCB, Canada Child Tax Benefit (CCTB) and National Child Benefit Supplement (NCBS) in 2016
    - More targeted to low and middle income families
- Research (Schirle, 2015) shows that UCCB reduced labour supply of women with children under 6
  - By about 3.2 percentage points for low-educated married women

Welfare

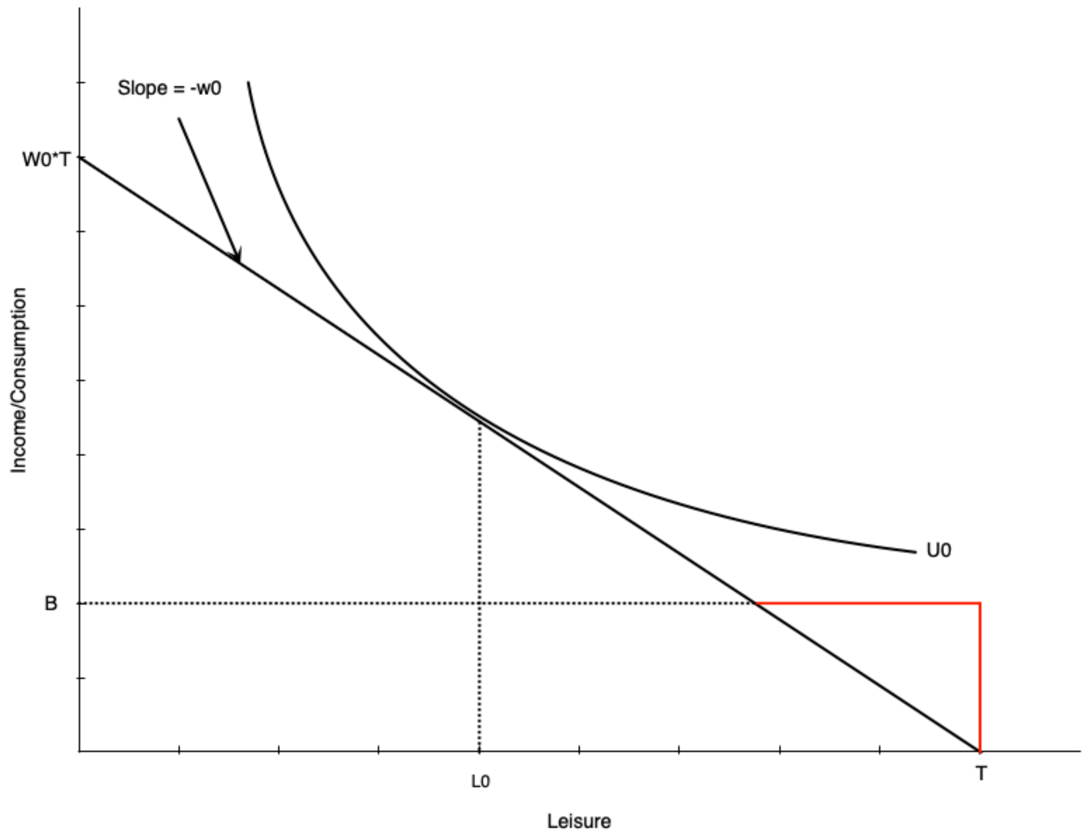
- Welfare is a lump-sum payment to non-workers that is reduced dollar for dollar with earned income
  - People do not really use the term welfare in Canada
  - Usually called social assistance or income assistance
- These programs have been criticized historically for negative work incentives
  - Creates incentives for some not to work
  - Important to balance this against the need to provide income support

Welfare



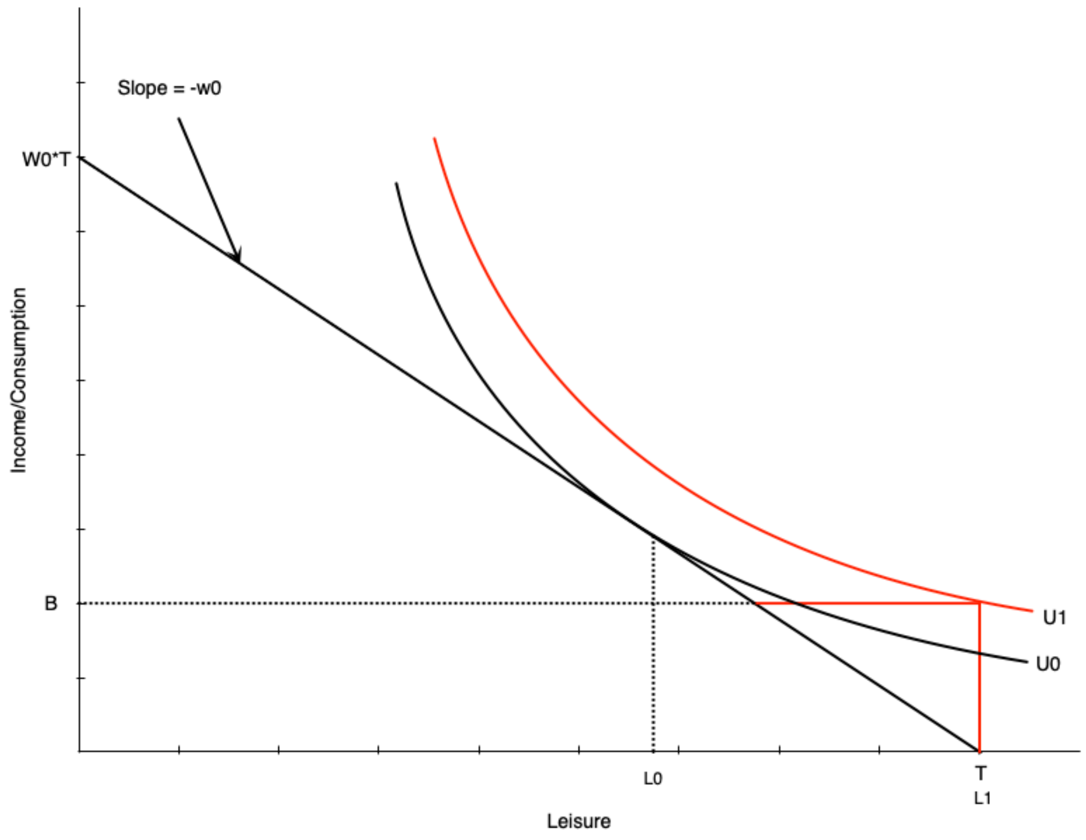
- Budget line is initially the black line
  - Leisure optimized at L_0 hours
- With welfare benefit is initially red line, then black line after they intersect
  - Welfare benefit B at zero hours of work
  - Benefit reduced by one dollar for each one dollar earned
    - Like a 100% tax on earned income
  - Income stays constant at welfare benefit until earned income exactly equals benefit
  - After that, person earns wage w_0 for each hour worked

Welfare



- For some people welfare has no effect on work hours
- In this graph, person is working H_0 hours prior to benefit
- After benefit, still maximize utility at H_0 hours
- Happens generally with people already working many hours

Welfare



- Others may face work disincentive
- A person like the left graph would choose to work zero hours after the welfare benefit
- Would happen because
  - Welfare benefit is generous
  - Individual puts high value on leisure
- This is not an ideal effect of welfare programs

Welfare

- How to combat work disincentive?
  - Reduce benefit level
    - But may not provide enough income support
  - Increase wages
    - Through minimum wages, subsidies, job training, unions
    - Can be expensive, which might reduce labour demand
  - Change preferences
    - Alter personal feelings towards work/leisure
    - Hard to do
  - Negative income tax
    - Reduce the tax back rate of extra earnings
    - Discussed in detail in a few slides

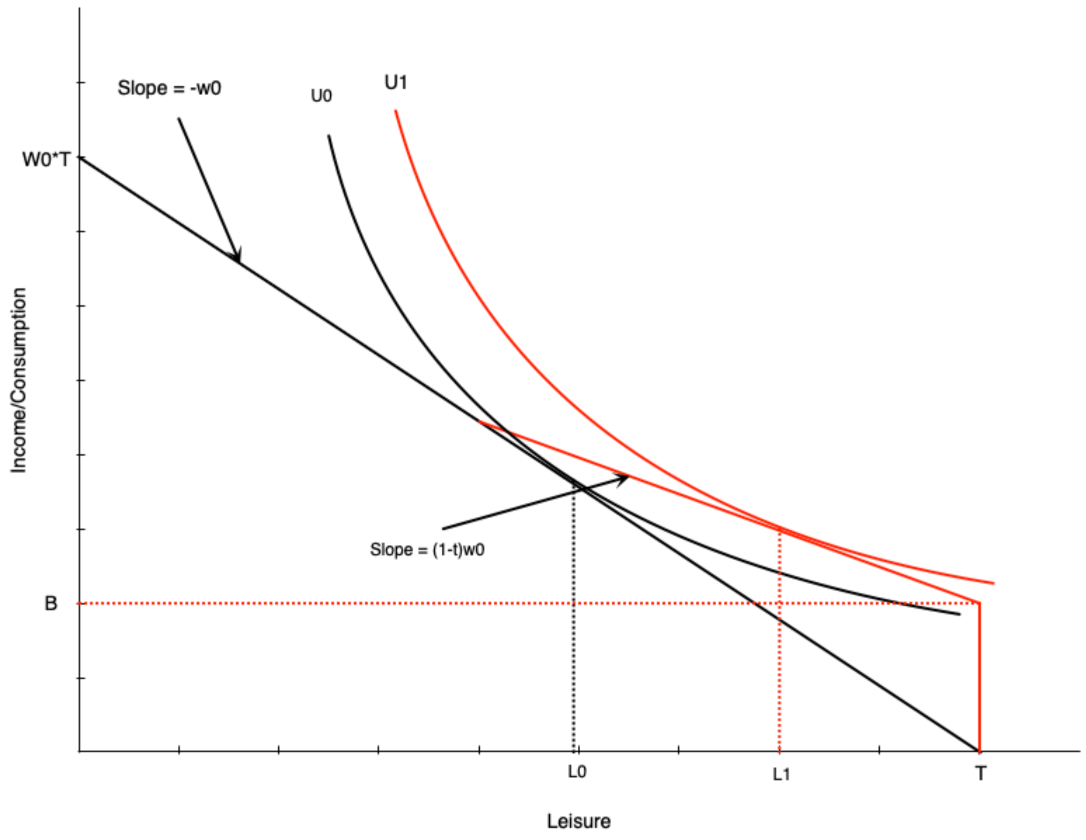
Welfare

- Some evidence comes from a policy change in Quebec
- Until 1989 welfare benefits for singles or couples with no kids was based on age
  - Under 30: $185/month
  - 30 and over: $507/month
- Turning 30 is associated with a big jump in welfare benefits
- Compared to 29 year olds, 30 year olds were
  - Spending much longer on welfare
  - Less likely to be employed

Negative Income Tax

- Negative Income Tax (NIT) is a welfare program where the benefit is reduced at a rate less than one dollar for each dollar earned
  - For example, a benefit of $10,000 that is reduced by 50 cents for each dollar earned
- Idea is to provide income support with fewer work disincentives
  - Because the benefit is reduced at a lower rate, there is still an incentive to work
- Same as a Guaranteed Annual Income (GAI) or Basic Income
  - A minimum income level guaranteed by the government

Negative Income Tax



- Negative Income Tax (NIT) has two components
  - A benefit B at zero hours of work
  - A tax rate t on earned income until earned income exactly equals benefit
- This pivots the budget line up relative to pure welfare scheme
- Individual will reduce work hours from H_0 to H_1
- Work disincentive is less than pure welfare case
  - This person would drop out of the labour force with pure welfare, but still works some hours with NIT

Negative Income Tax

- Introducing NIT has income and substitution effects
  - Income effects
    - The benefit increases non-labour income, so increases leisure
    - The tax has a negative income effect, so reduces leisure
    - On net, income effect of benefit will dominate
  - Substitution effect
    - Because of the tax rate, the effective wage is lower
    - This makes leisure relatively cheaper, which reduces work hours
- Overall effect on work hours is negative, but smaller than pure welfare case

Negative Income Tax

- Mathematically, the budget constraint with the NIT is

$$C = \begin{cases} wh + B - twh, & \text{if } B - twh \geq 0 \\ wh, & \text{otherwise} \end{cases}$$

- Says that consumption is
  - Earned income (wh) plus net benefit ($B - twh$) if net benefit is positive
  - Just earned income (wh) if the benefit is zero or negative
- The point at which the benefit is zero is called the break-even point
  - Occurs when $wh = \frac{B}{t}$

Negative Income Tax

- The province of Manitoba implemented a NIT program called MINCOME in the 1970s
- Randomized controlled trial
  - Some people were given NIT (treatment), others not (control)
  - Randomly allocated to treatment and control groups
- Conducted in the cities of Dauphin and Winnipeg
  - Dauphin is a small town, Winnipeg is a large city
  - Dauphin was saturated, meaning everyone got NIT
- There were 7 treatments with different benefit levels and tax rates
- Program was short lived, and ended in 1979 due to political changes

Negative Income Tax

Guarantee at enrolment	Tax Rate (on total Income)		
	35%	50%	75%
$3800	Plan 1	Plan 3	*
$4800	Plan 2	Plan 4	Plan 7
$5480	X	Plan 5	Plan 8

Plan 9

* Plan 6 was collapsed into Plan 7 because participants viewed this option as too punitive and left the experiment.
X This option was deemed to expensive for the budget.
Plan 9 was the control group

MINCOME Treatment Groups. Source: Wikipedia

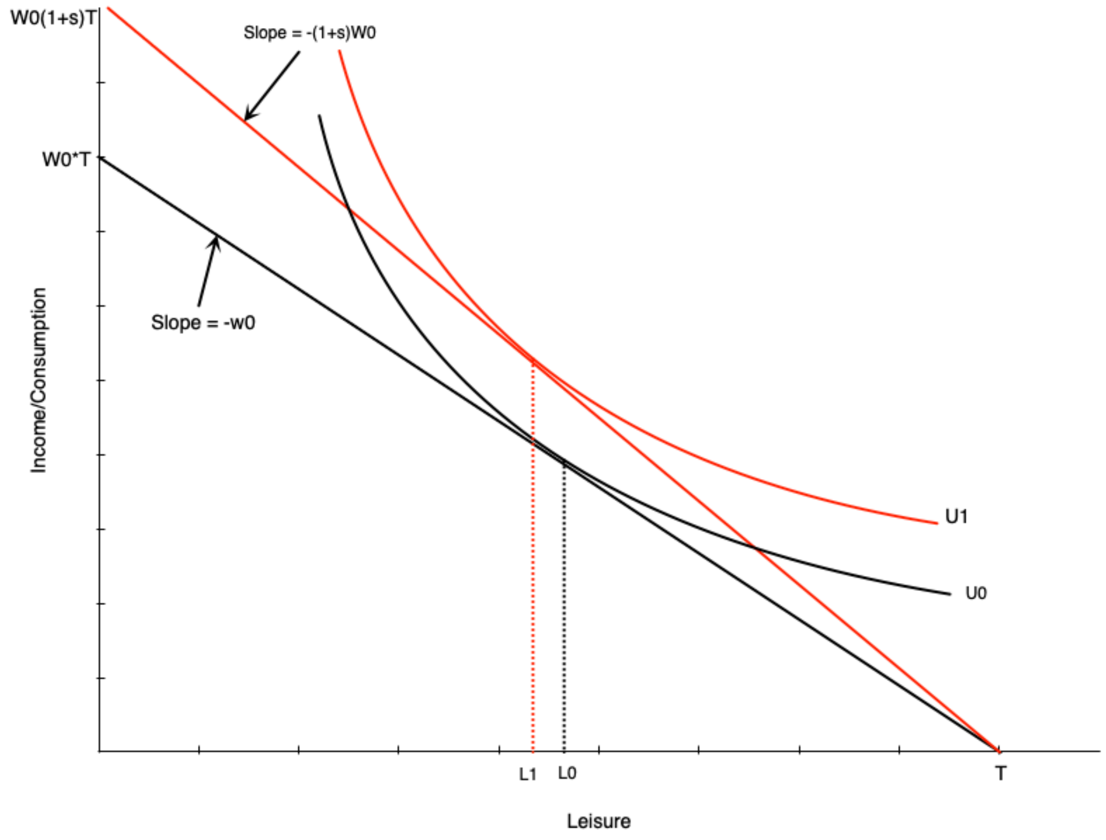
Negative Income Tax

- Results show small decline in hours worked
  - Mainly by mothers of newborns and women in two-parent families
  - No significant decline for men
- But significant improvements in health outcomes
  - Fewer hospitalizations and doctor visits
  - Especially for mental health and accidents
- Increase in adolescents finishing high school

Wage Subsidies

- A **wage subsidy** is a payment to workers that increases their wage rate
  - For example, a subsidy of $5/hour increases a wage of $15/hour to $20/hour
- This simply increases the wage rate
  - Budget line pivots up
- There are both income and substitution effects
  - Income effect increases leisure
  - Substitution effect decreases leisure
- Overall effect on work hours is ambiguous

Wage Subsidies



- Budget line pivots up with wage subsidy
- Graph shows person working more with subsidy
- But effect is ambiguous
  - Income and substitution effects work in opposite directions

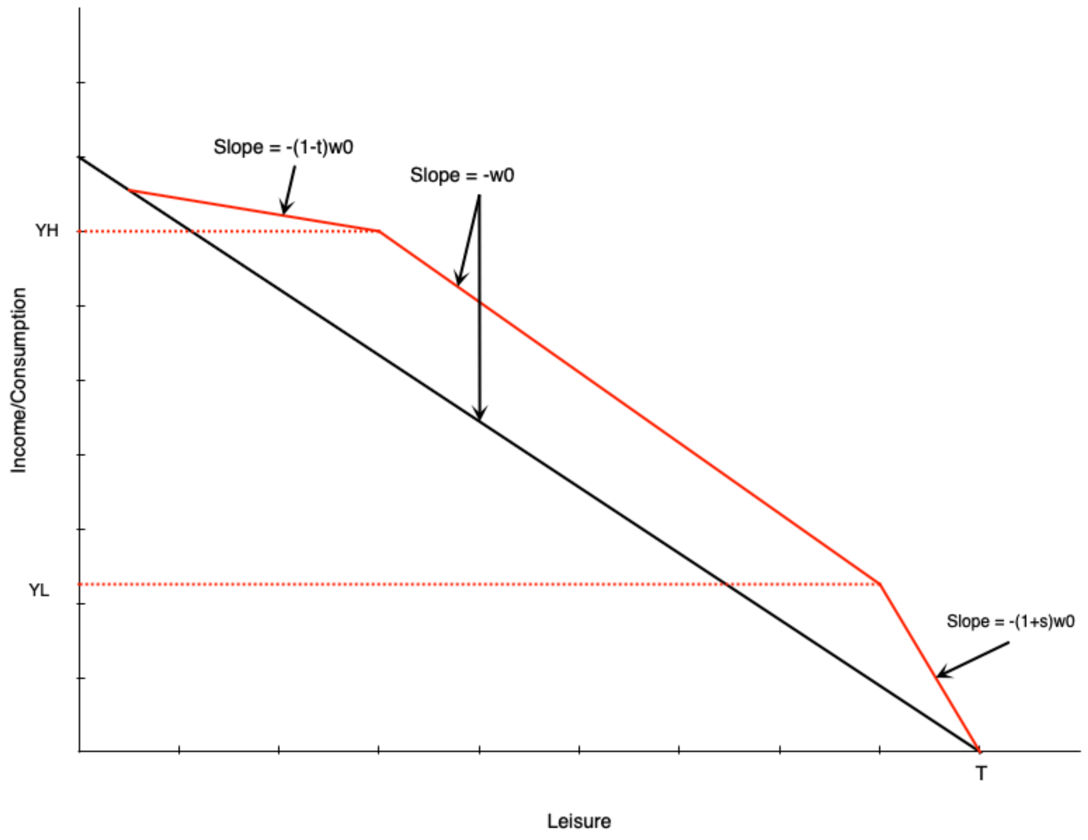
Wage Subsidies

- Wage subsidies have less negative work incentives than welfare or NIT
  - Because they do not reduce the effective wage rate
- But they do not provide income support to non-workers
  - So may not help those in most need
- Examples of pure wages subsidies in Canada mostly go to *employers*
  - Unclear if any of this gets passed on to workers

Tax Credits

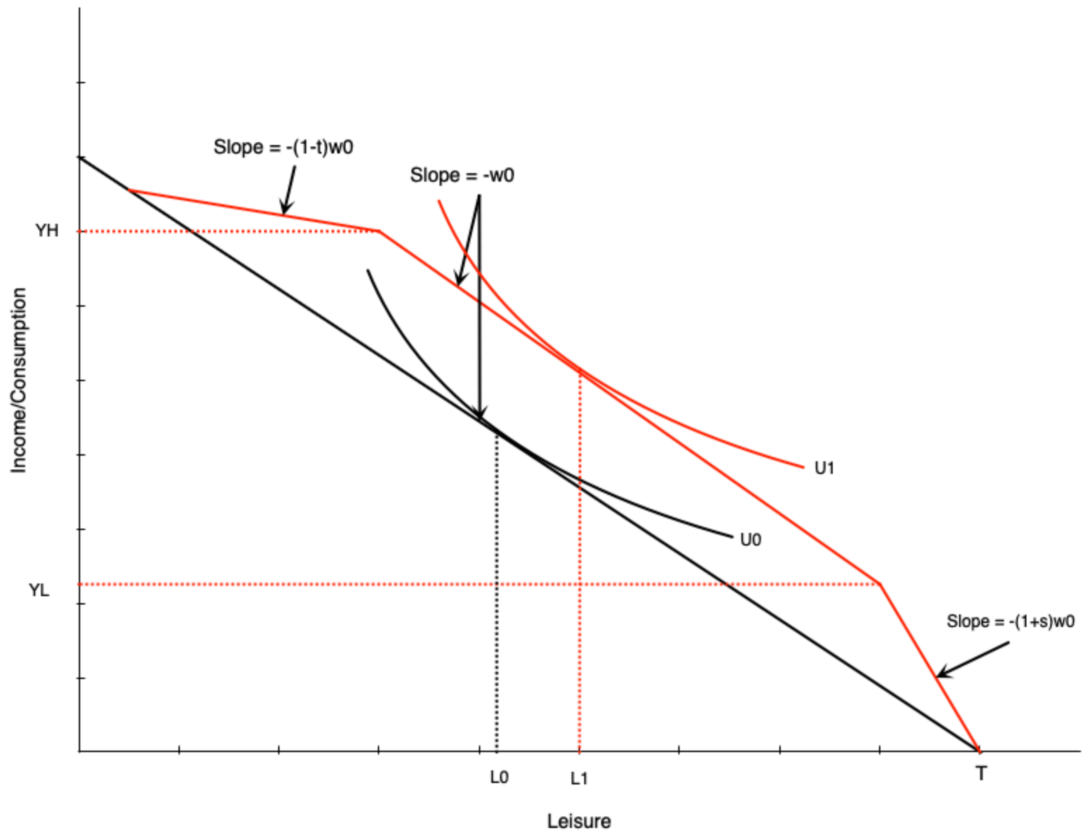
- Income support is sometimes provided through the tax credits
- A **tax credit** reduces the amount of tax owed, and there are two types
  - Non-refundable: can reduce tax owed to zero, but not below zero
  - Refundable tax: can reduce tax owed below zero, resulting in a payment to the individual
- Some income maintenance programs are delivered through refundable tax credits
  - Canada Worker Benefit (CWB) in Canada
  - Earned Income Tax Credit (EITC) in the US
- They have a similar structure to the following graph

Tax Credits



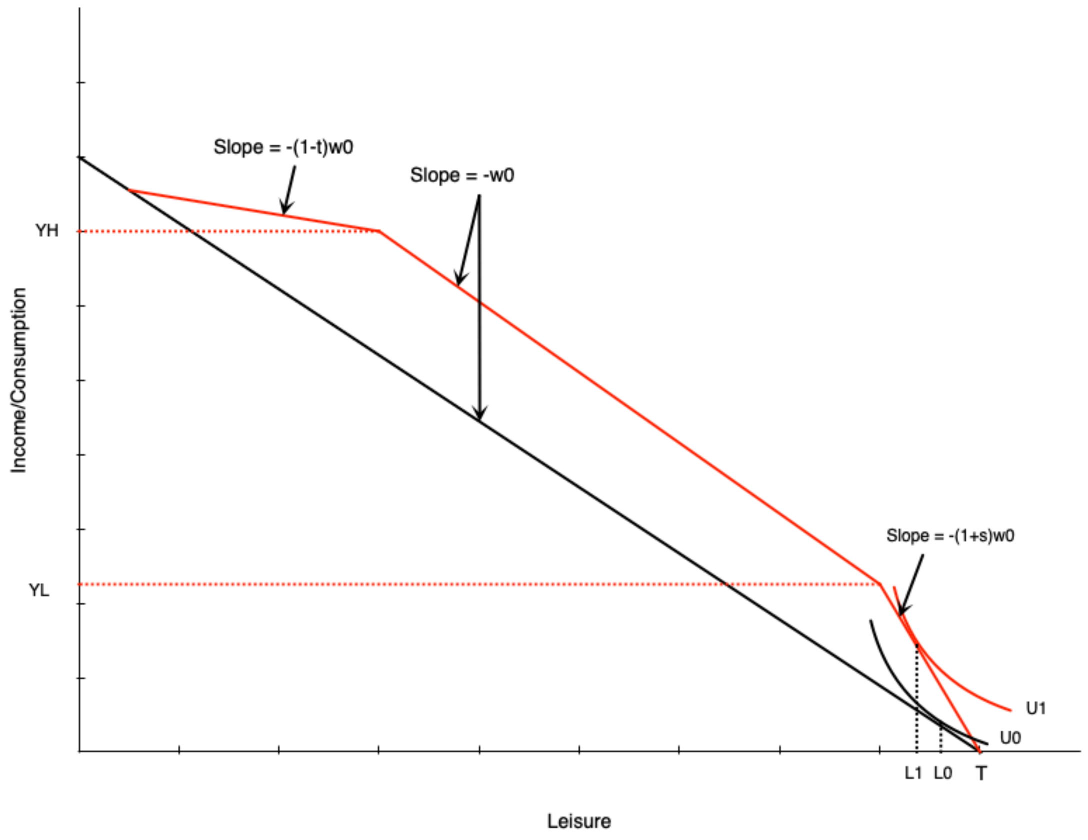
- Budget line shows program similar to CWB or EITC
- Has three segments
  - Phase-in: subsidy proportional to earned income
  - Plateau: maximum benefit
  - Phase-out: benefit reduced with earned income
- Has elements of wage subsidy, demogrant, and NIT

Tax Credits



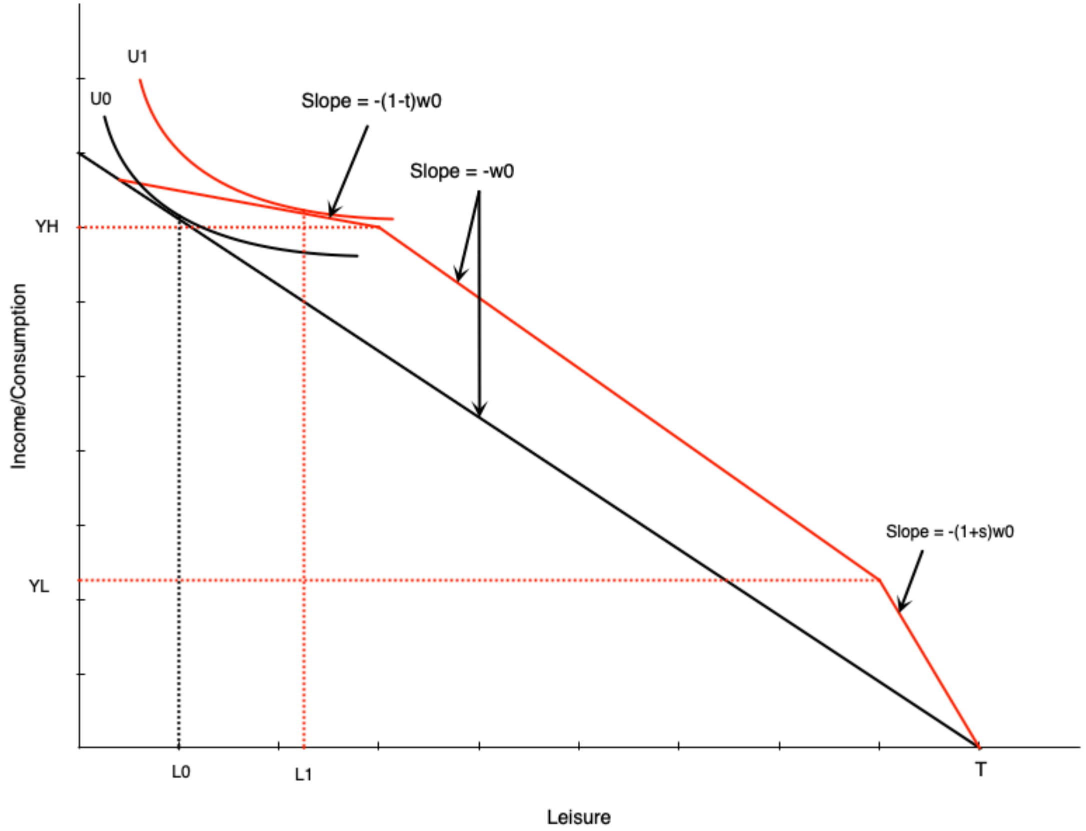
- Picture someone initially in middle range
- Tax credit gives them maximum benefit
- Acts like pure income effect
- If leisure is a normal good, they will reduce work hours

Tax Credits



- Someone in the phase-in range will see an increase in the wage
- Acts like a pure wage subsidy
- Has income and substitution effects that move in opposite directions
- Effect on hours is ambiguous

Tax Credits



- Finally imagine someone in the phase-out range
- This is like a negative income tax
- Income and substitution effects both reduce work hours
- Effect on work is unambiguously negative

Unemployment Insurance

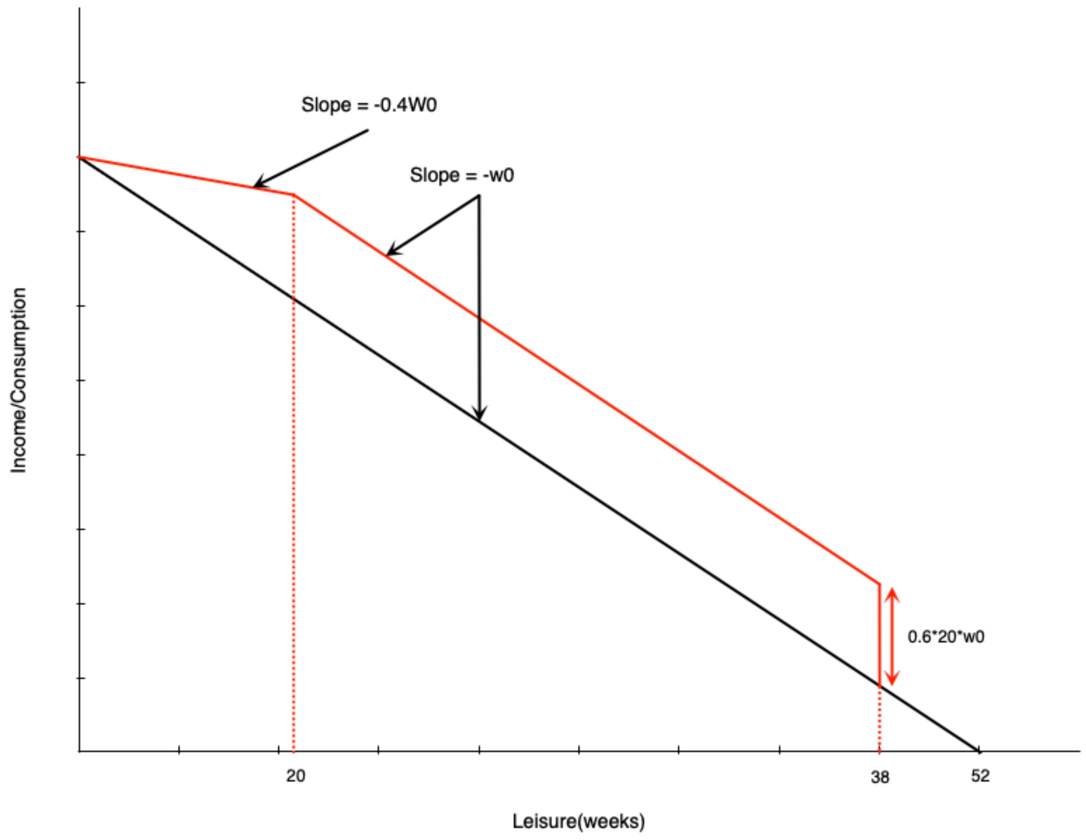
Introduction

- Unemployment insurance schemes create a safety net for workers who lose their job
  - Collect income for a period while not working
- In Canada, this is called Employment Insurance (EI)
  - Funded by payroll taxes on employers and employees
- The EI program has two main components
  - Regular benefits for those unemployed through no fault of their own
  - Special benefits for sickness, maternity, parental leave, caregiving
- For regular benefits you could get
  - 55% of your average insurable weekly earnings, up to a maximum amount
  - Benefits last between 14 and 45 weeks depending on unemployment rate and hours worked in last year

Introduction

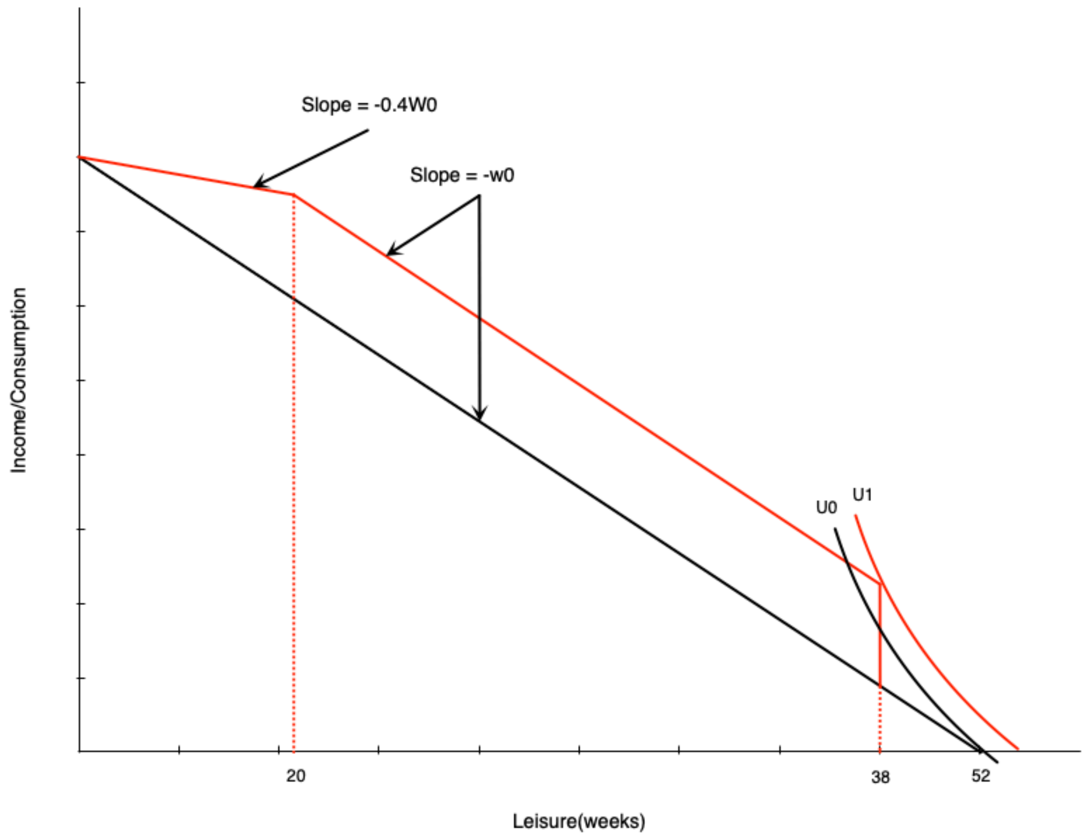
- We can analyze EI with the income-leisure diagram
- To do this, simplify the program to have two parameters
  - Must work 14 weeks in the year to qualify
  - Benefit is 60% of weekly wage up to a maximum of 20 weeks
- Person can therefore work up to 32 weeks in the year and get full benefits
  - For each week worked beyond 32, the lose benefits for that week
  - Because benefits are 60% of wage, this is like a 60% tax on earned income beyond 32 weeks
- Cannot work and get EI simultaneously

EI Budget Constraint



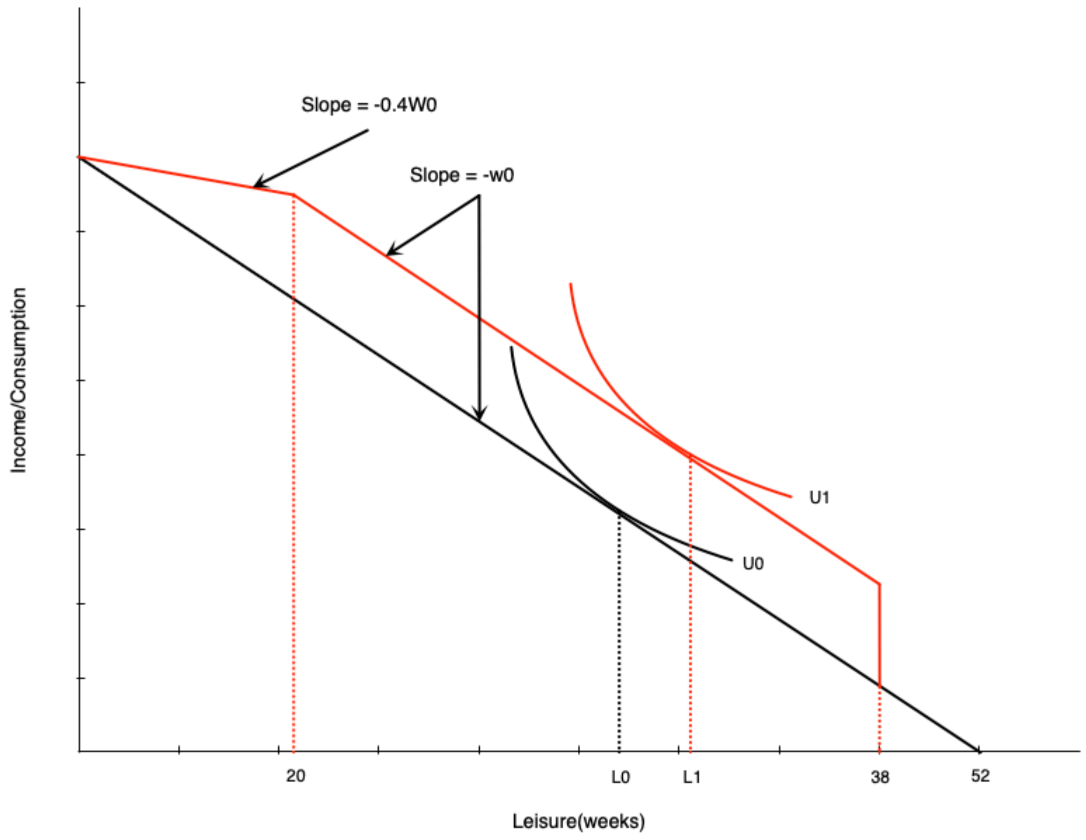
- First 14 weeks are paid at usual wage w_0
- If person consumes 20-38 weeks of leisure (works 14 to 32 weeks), gets 20 weeks of benefits
  - Benefit is 60% of 20 weeks of wage, or $.6 \times 20 \times w_0$
- Less than 20 weeks of leisure (more than 32 weeks of work) reduces amount of benefit week for week
  - Benefits are paid at 60% of wage
  - Work is paid at 100% of wage
  - So person keeps 40% of wage in this range

EI Optimum



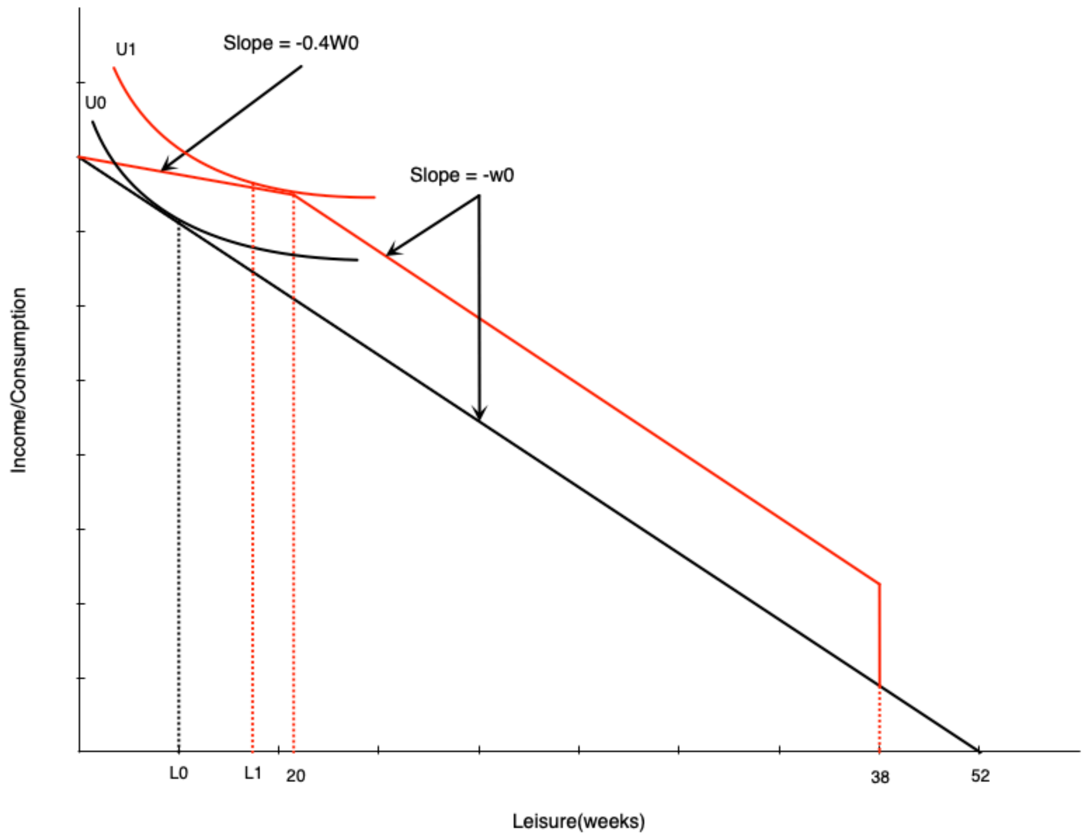
- Behaviour depends on position on budget line
- A non-participant might start working to get the benefit
- Happens because large increase in non-labour income happens after 14 hours of work

EI Optimum



- In the middle portion of the budget line, the person is getting full benefits
- This is like a pure income effect, so they will reduce work weeks

EI Optimum



- On the left, it again acts like a negative income tax
- Income and substitution effects both reduce work hours
- Effect on work is unambiguously negative

Evidence on EI and Labour Supply

- Maine vs New Brunswick
- Initially, had very similar employment insurance schemes
- Later, New Brunswick increased generosity of benefits relative to Maine
- Analyze with a difference-in-differences approach
  - Compare changes in labour supply in New Brunswick before and after change to Maine over same period
  - Maine acts as a control group for the change in New Brunswick
    - Cannot look just at New Brunswick over time because other factors may be changing
    - Changes in Maine over time capture and control for the change in these other factors

Evidence on EI and Labour Supply

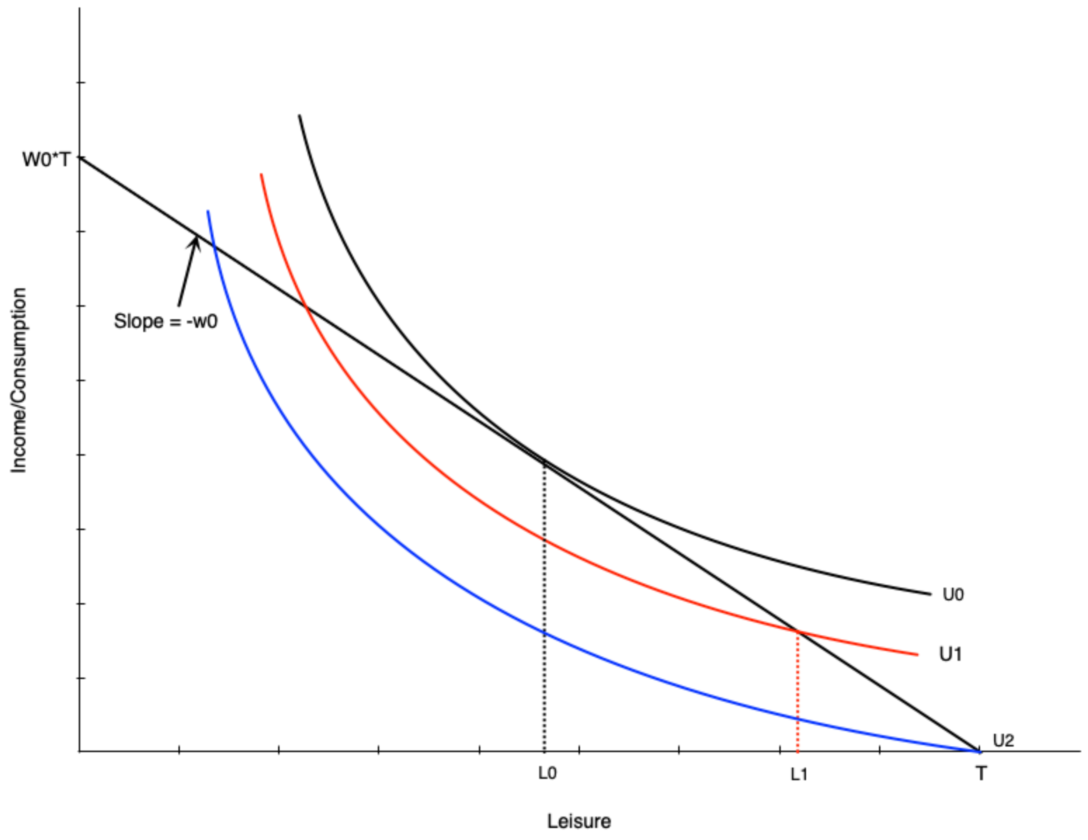
- Analysis finds
  - Higher labour force participation in NB for marginally-attached workers
  - Lower hours for those with stronger attachments
- This corresponds exactly to the model predictions
- Highlights work disincentive effects of income maintenance programs

Disability and Workers Compensation

Introduction

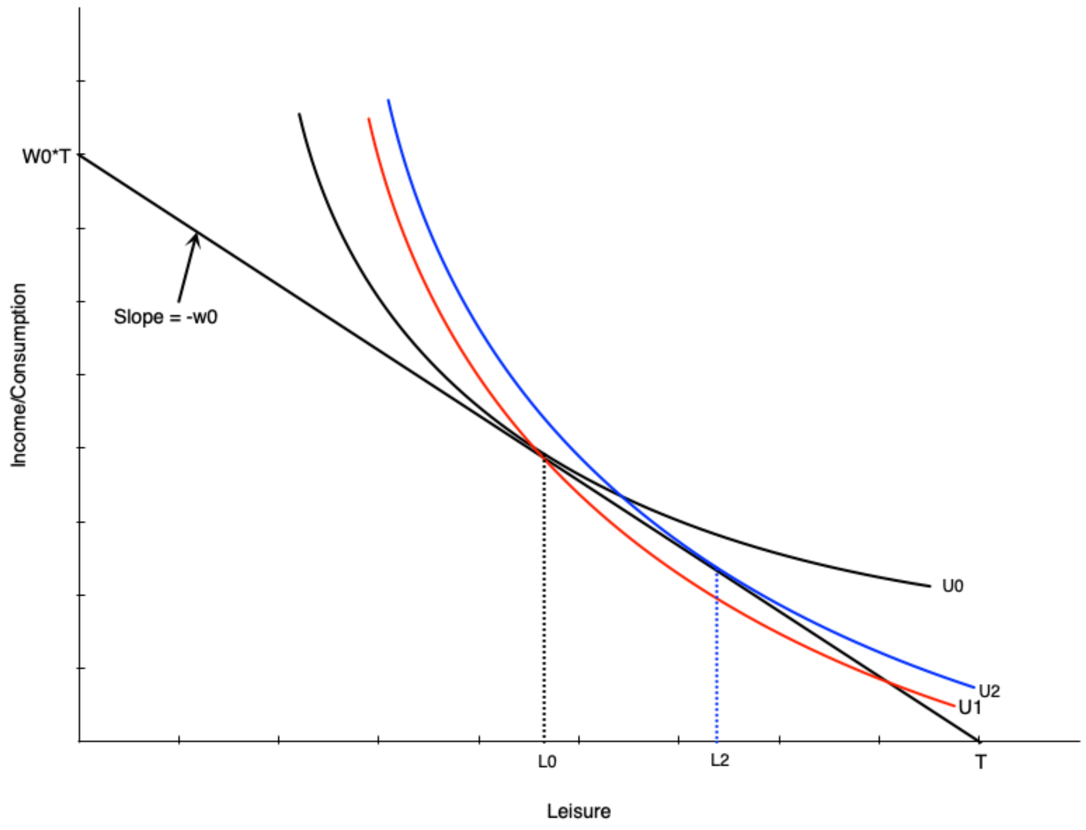
- Disability insurance and workers compensation provide income support to workers who are injured or become ill
  - Can come from public and private sources
  - Premiums usually paid by both employers and employees
- In cases of sever injury or disability work incentives are not a concern
  - People are unable to work
- In others people may still be able to work

Effect of a Disability on Labour Supply



- Initially person chooses L_0 leisure and works H_0 hours
- If they become injured but can still work, the might choose L_1 leisure and work H_1 hours
  - Now on a lower, non-optimal indifference curve
- If they become completely disabled, they choose T leisure and work zero hours
  - On an even lower indifference curve

Effect of a Disability on Labour Supply

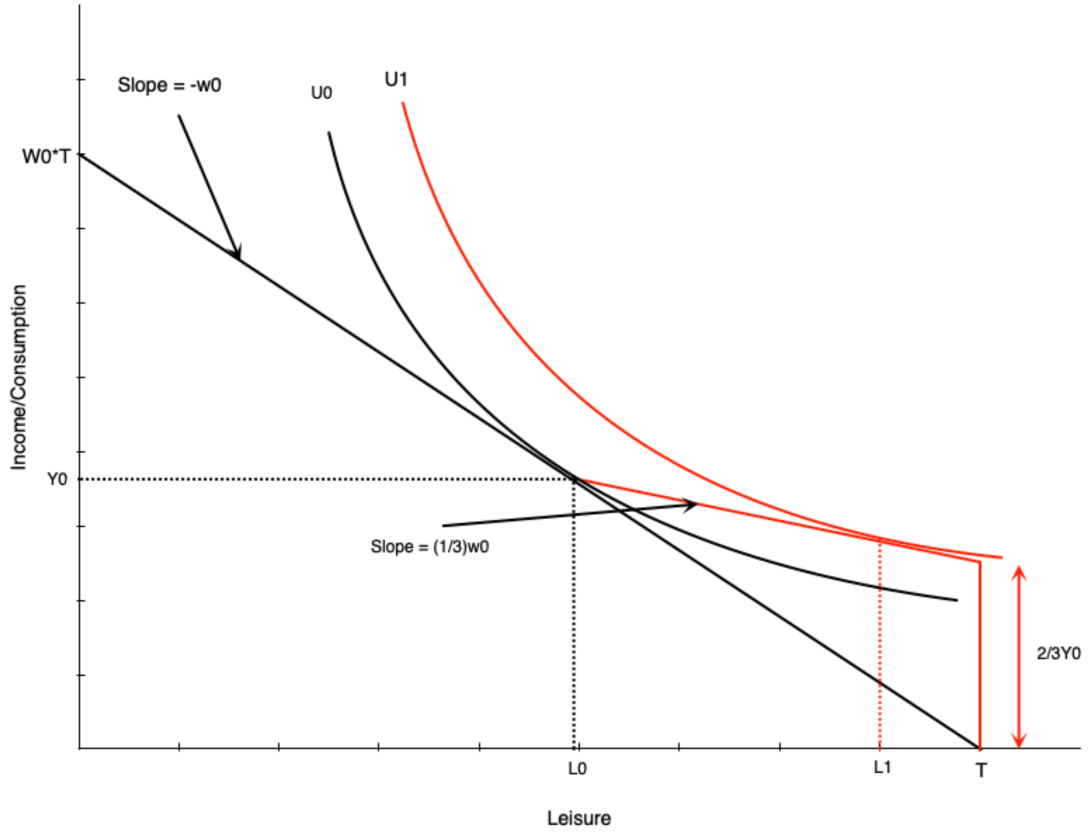


- A disability might also change their preferences
- If work becomes more difficult, preferences would tilt towards leisure
  - Indifference curve becomes steeper
  - MRS increases at each point
  - More willing to give up consumption to get leisure
- Even if the person can still work unconstrained, they will choose more leisure
  - Person would go from L_0 to L_2

Effect of Compensation

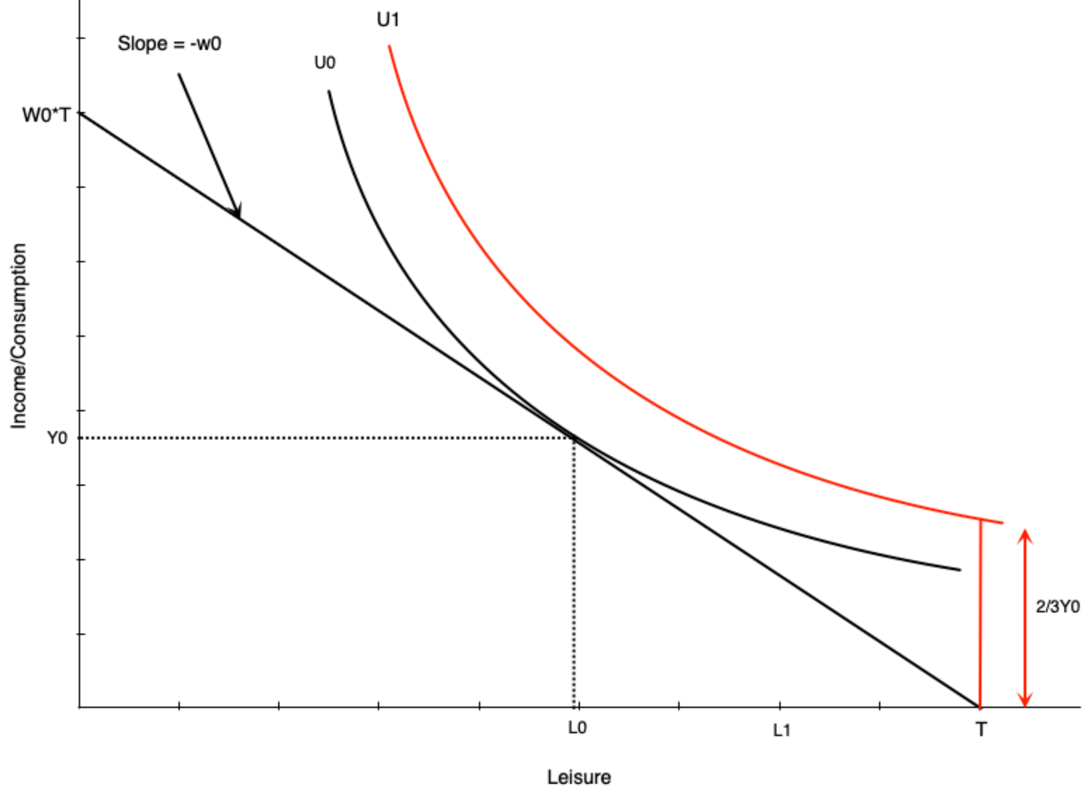
- If a person now receives disability insurance, their budget constraint changes
- These programs usually pay a fraction of previous earnings
  - In Ontario, the Workplace Safety and Insurance Board (WSIB) pays 85% of net earnings (up to a max)
  - Lasts until you can return to work or age 65
    - Another program would take over at that point
- Below we examine two schemes
  - An income replacement of two-thirds
  - Full income replacement

Effect of Compensation



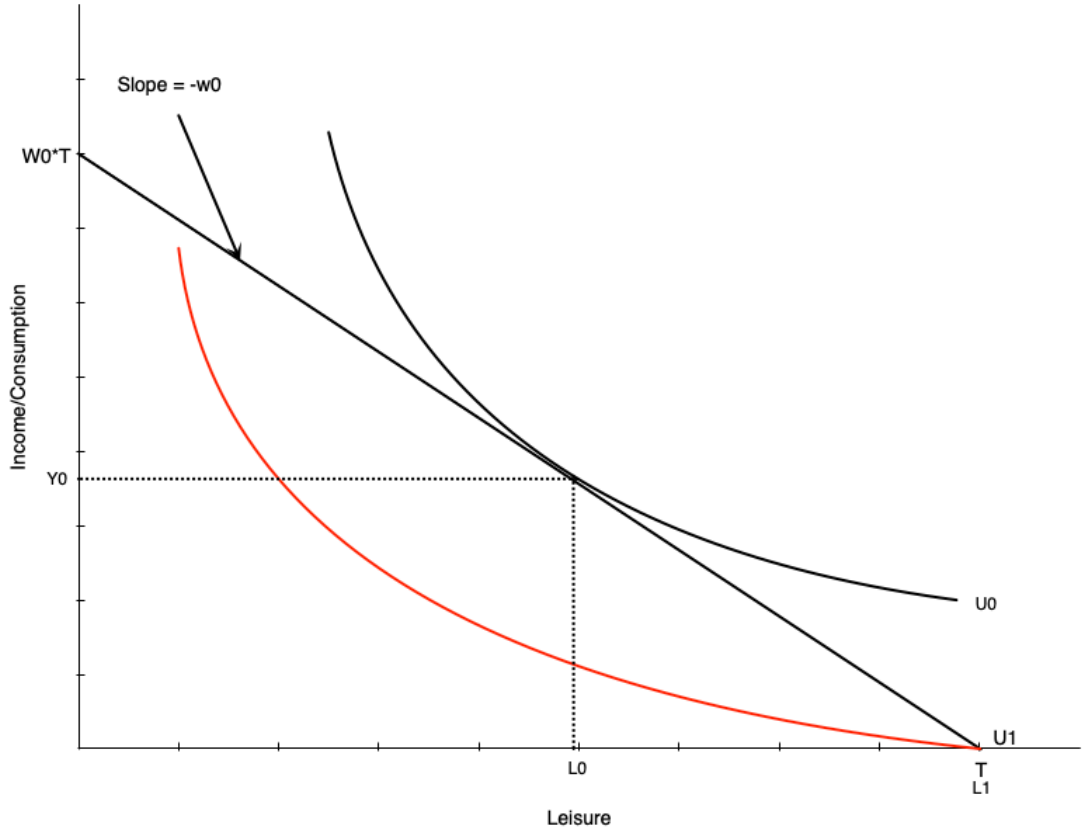
- A two thirds compensation scheme looks like a negative income tax
- At zero hours of work, person gets two-thirds of previous income
- For each hour of work beyond zero
  - They gain their full wage
  - But lose the benefit (two thirds of wage)
  - So they keep one third of their wage
- For workers that can chooose any level or work hours, effects on labour supply match NIT
  - Negative effects on hours

Effect of Compensation



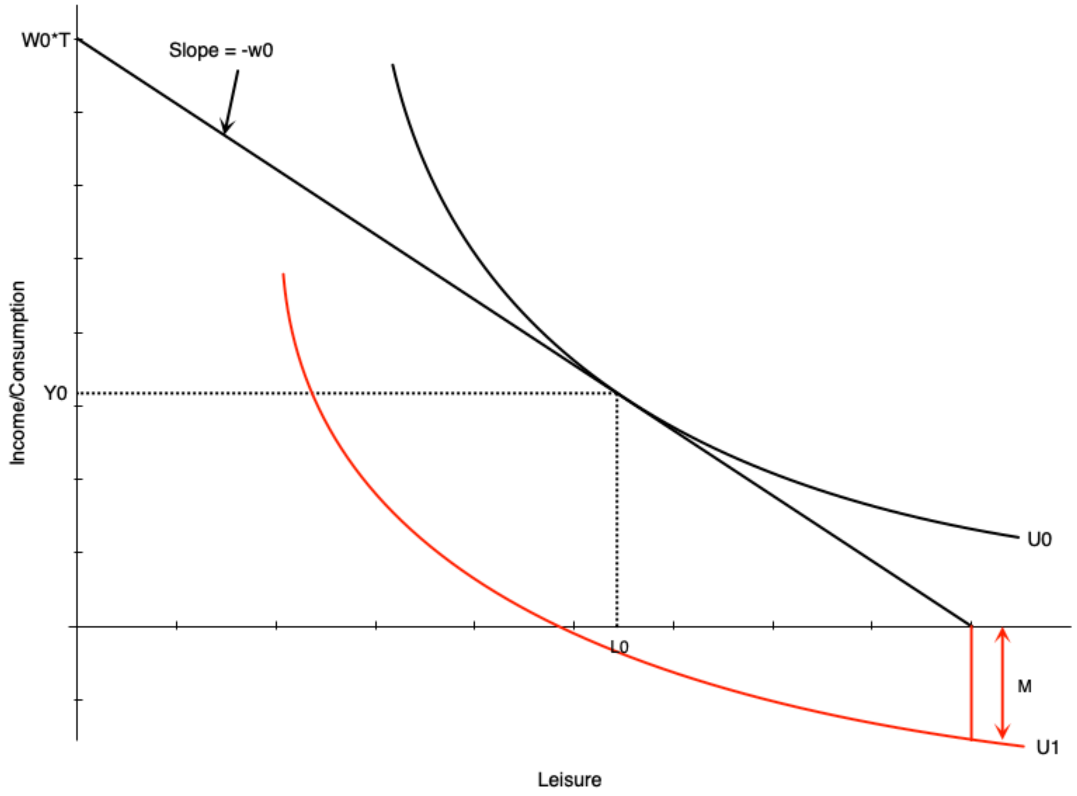
- What if the worker cannot choose any level of work hours?
- Imagine they can only work H_0 hours or zero hours
- If scheme is generous enough, person chooses zero hours
  - Utility at zero hours is higher than at H_0 hours
- Strong work disincentives of such a scheme

Effect of Compensation



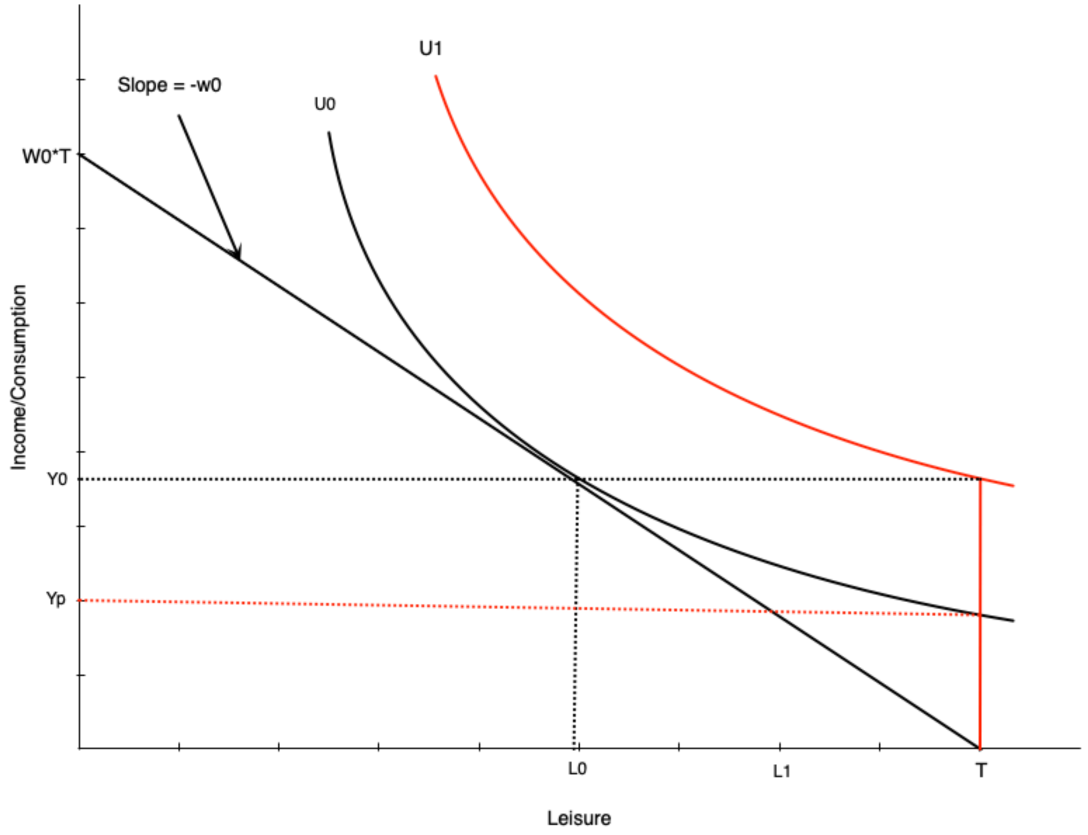
- Not providing any compensation could be problematic
- A person who can work would choose to work H_0 hours
- If the person cannot work, they get lower utility
  - End up substantially worse off

Effect of Compensation



- Things get worse if there are medical expenses
- These expenses act like a reduction in non-labour income
- If large enough, non-labour income could go negative
  - Would happen with non workers compensation or very low compensation
- This is an extreme scenario

Effect of Compensation



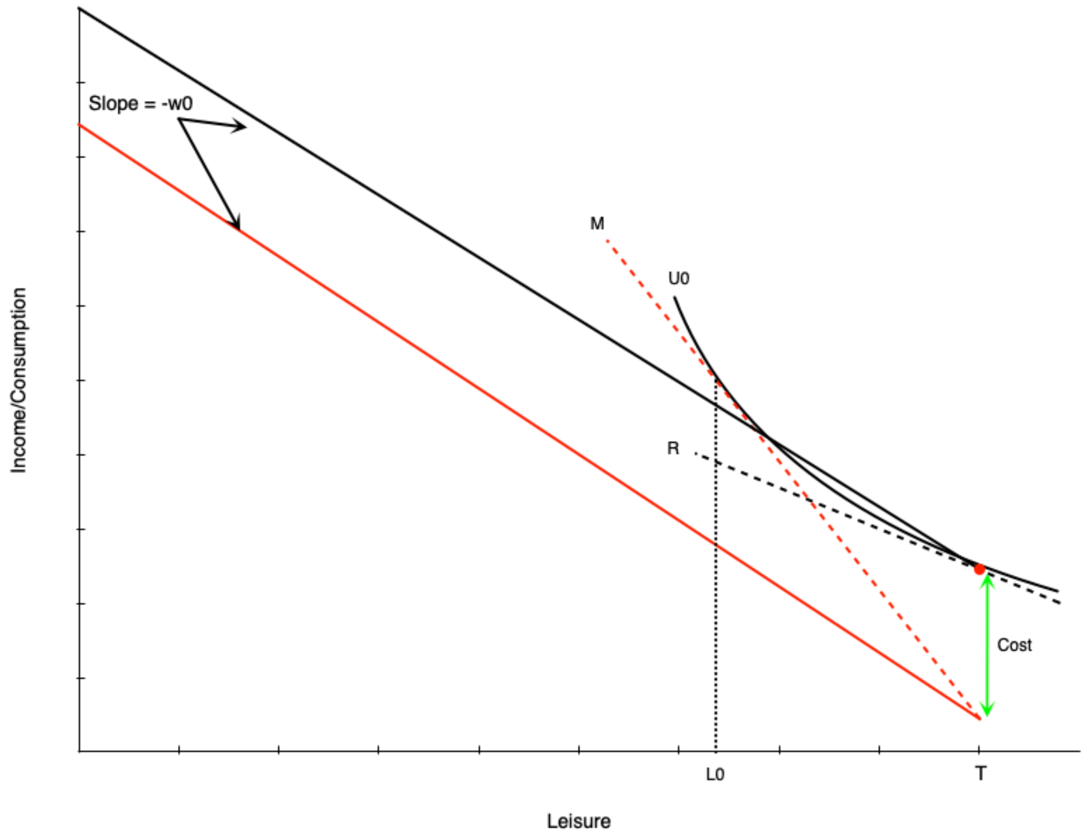
- How much compensation should we provide to someone who does not work?
- One option is full income replacement
  - Would put the person on a higher indifference curve (U_1) than working
- Could also maintain the same satisfaction level as working
  - Would put the person on the same indifference curve as working
  - Lower compensation than full income replacement (Y_p)
- The optimal level is a policy question

Child Care Subsidies

Introduction

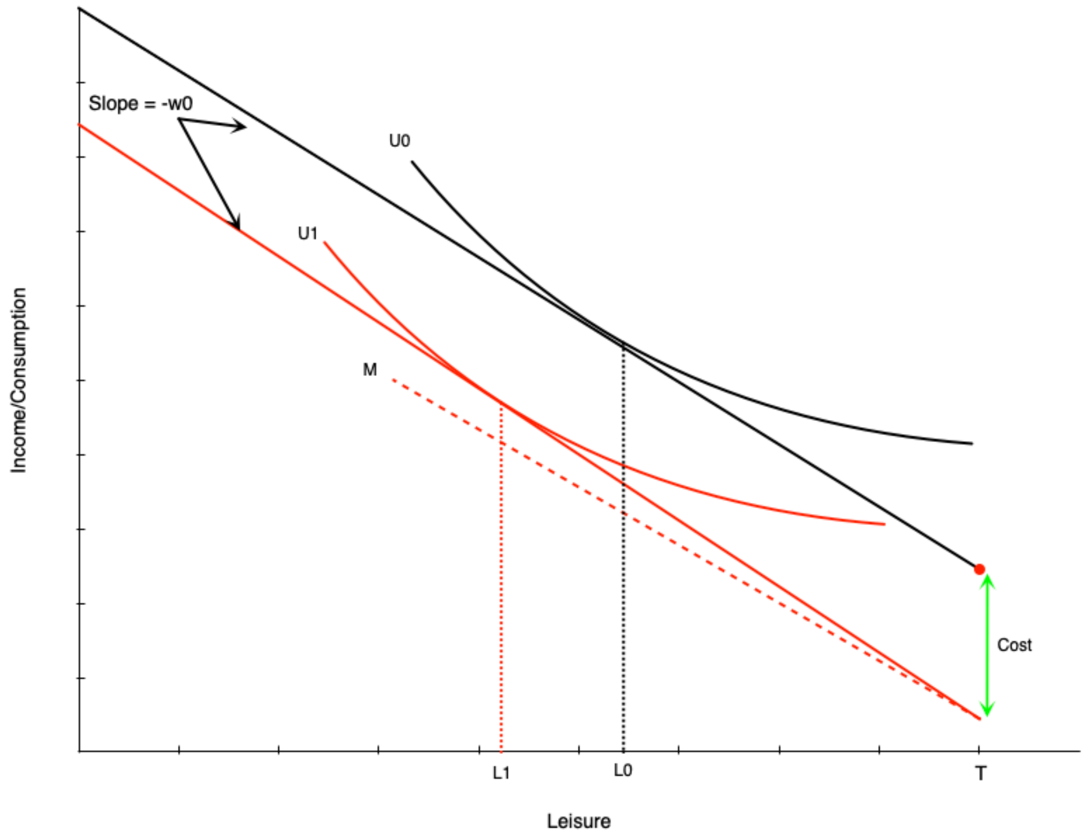
- Child care subsidies are payments to parents to help cover the cost of child care
- Canada is currently expanding child care subsidies
  - Goal is to reduce average cost of child care to $10/day by 2026
  - To date more than half of provinces have a $10/day average
  - In some provinces, subsidy is very substantial
  - Complicated because agreements must be reached with provinces
- Quebec has had a $10/day child care program since 1997
  - Provides a useful case study of the effects of child care subsidies
- Here we study how this affects labour supply

Child Care Costs



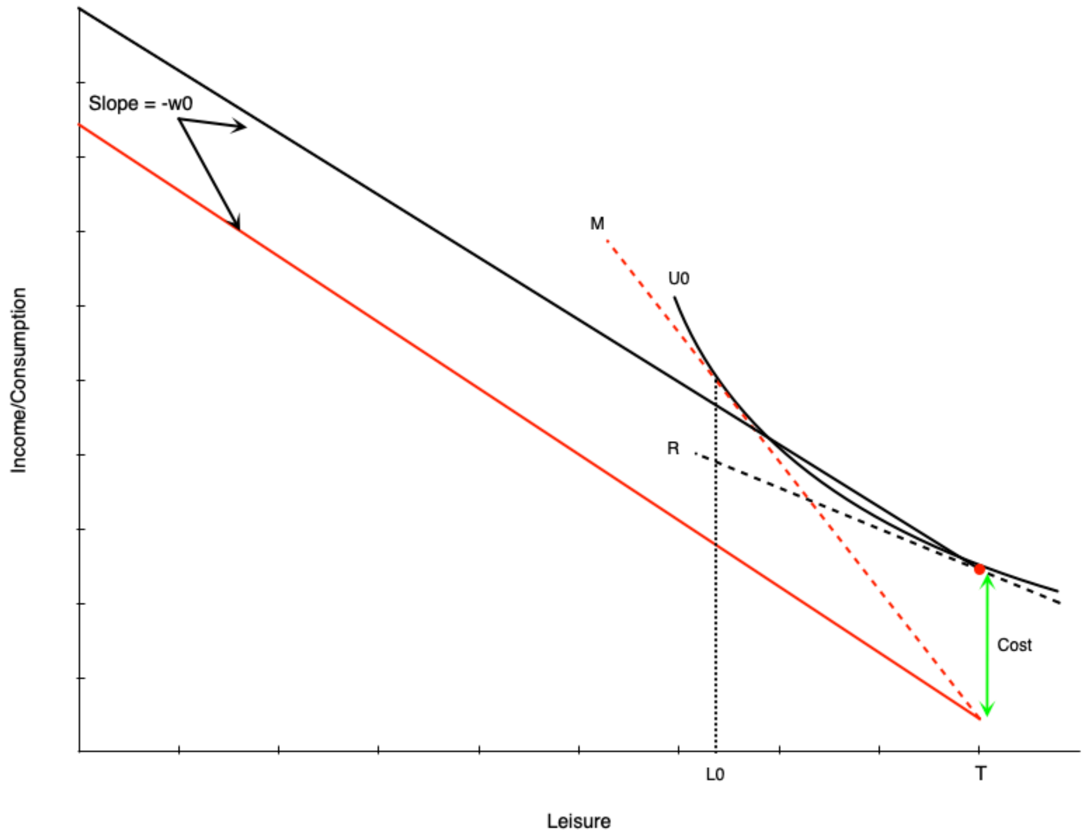
- With child care costs the budget line becomes discontinuous
  - At zero hours of work, it is red dot
  - For any amount of work, it drops down by the amount of child care costs
  - Then is a straight line with slope equal to wage rate
- Child care costs increase reservation wage
  - Due to costs, person needs higher wage to induce them to work
- In this graph the person chooses not to work

Child Care Costs



- A person working would choose more hours
- This person has reservation wage above the wage rate with child car are costs, so works positive hours
- Child care costs act as a negative pure income effect
  - Choose less leisure, more work
- Must work more to make up income loss from child care costs

Child Care Costs



- Child care costs create a gap in the labour supply schedule
- A person who chooses to work will work a minimum of $H_0 = T - L_0$ hours
  - Not worth it to work less than this amount
  - Due to large child care costs
- This means labour supply schedule jumps from zero hours to H_0 hours

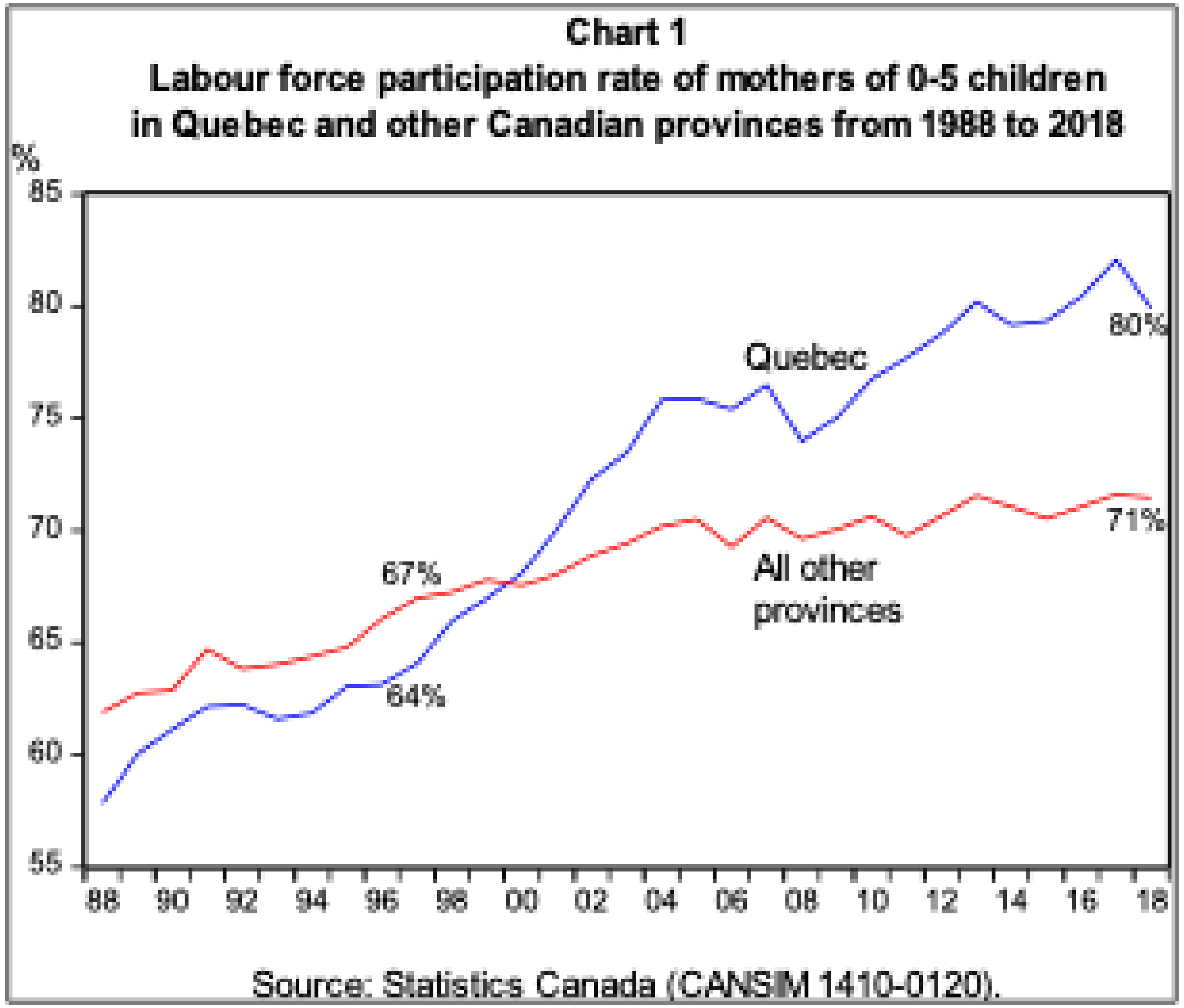
Child Care Subsidies

- A child care subsidy reduces the cost of child care
- For example, a full subsidy (driving costs to zero) would
  - Move the budget line up in parallel
  - Reduce the reservation wage
- The lower reservation wage would induce participation
  - Because barrier of child care is removed
  - This would encourage part time work in particular
- Shift up in the budget line would reduce work hours
  - Pure positive income effect
  - Increases consumption of leisure, reduce work hours

Evidence on Child Care Subsidies

- Quebec implemented a $5/day child care program in 2000
  - Now a sliding scale between $8.25 and $21.45 today
- Researchers used a difference in differences approach to analyze the effect
  - Compared changes in labour supply of mothers in Quebec to those in other provinces over same period
  - Other provinces act as a control group for changes in Quebec
- Found that labour force participation of mothers with young children increased by 7.7 percentage points
- Unfortunately also found negative effects on child mental health, non-cognitive behaviour, crime later in life

Evidence on Child Care Subsidies



Source: Fortin (2019)

Summary

Summary

- Income maintenance programs provide income support to various groups for various reasons
- Some of these programs can create work disincentives
  - Need to balance this against the need to provide income support
- We examined several different programs that exist today with labour supply model
- In next section, we look at labour supply over the life cycle